

Problem Set 3: Loops and Functions

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1 Exercise 1: Evaluating a Function

1. Define the mathematical function

$$f(x) = \frac{x+1}{x-1}.$$

The function has a pole at $x = 1$. Write a Python function

```
def f(x):  
    return (x + 1) / (x - 1)
```

Use the list

```
values = [0, 0.5, 0.9,  
          1.1, 1.5, 2]
```

and a `for` loop to evaluate and print x and $f(x)$ for every value in the list.

- Your loop must call `f(x)` rather than repeating the formula inside the loop.
- Explain why $x = 1$ is not included in `values`.
- Add the values 0.99 and 1.01. What happens to the magnitude and sign of $f(x)$ near the pole?

```
("Olivia", 20681003, 3),  
("Liam", 20681004, 4),  
("Sophia", 20681005, 2),  
("Ethan", 20681006, 0)  
]
```

Create the roster structure

```
algebra = ["Algebra", []]  
trig = ["Trigonometry", []]  
calculus = ["Calculus", []]  
math_courses = (algebra, trig, calculus)
```

Use one `for` loop to process every student. For each student:

- extract the name, student ID, and score;
- call `placement(score)`;
- append the tuple `(name, student_id)` to the correct roster.

Do not repeat the placement rules inside the loop; the decision must be made by the function.

Finally, print `math_courses` and verify that every student appears exactly once and in the correct course.

2 Exercise 2: Placement Function

1. A placement score determines the appropriate math course:

Score	Course
0-1	Algebra
2-3	Trigonometry
4-5	Calculus

Write a function

```
def placement(score):  
    # use if, elif, else  
    # return a course name
```

that **returns** the appropriate course name as a string. Test the function first with

```
print(placement(1))  
print(placement(3))  
print(placement(5))
```

The results should be Algebra, Trigonometry, and Calculus.

Now use

```
students = [  
    ("Maya", 20681001, 5),  
    ("Noah", 20681002, 1),
```

3 Exercise 3: While-Loop Count

1. Use a simple `while` loop to count the total number of students stored in the three rosters.

Begin with

```
i = 0  
total = 0
```

and use `i` to access each element of `math_courses`. During each pass, add the length of that course's student list to `total`, then increase `i` by one.

Print `total`. What value should the program produce, and why?